

# Anthony Lu

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## Education

### New York University

New York, NY

B.A. Computer and Data Science, Minor in Mathematics - **GPA 3.76/4.00**

Expected May 2027

• **Coursework:** Deep Learning & LLMs, Machine Learning, Optimization for ML, Parallel Computing, Computer Systems, Algorithms, Data Structures, Linear Algebra, Probability & Statistics, Calculus III, Differential Equations.

## Skills

**Languages:** Python, JavaScript, TypeScript, Java, SQL, C/C++, Go, HTML/CSS  
**Frameworks:** React, Angular, Node.js, Next.js, FastAPI  
**Cloud/DevOps:** Google Cloud, AWS, Docker, Git, Perforce  
**ML/Data:** PyTorch, scikit-learn, Weights & Biases, Pandas, NumPy, HuggingFace

## Experience

### InterSystems

Boston, MA

*Software Engineer Intern*

May 2026 - August 2026

- Developing an agentic AI assistant for the InterSystems DTL Editor, combining compiler feedback and multi-step validation pipelines to generate and iteratively refine DTL transformations.
- Designing an MCP server that exposes DTL authoring, validation, and compilation tools for integrated and third-party agents.
- Implementing synthetic data and benchmarking pipelines to evaluate agent performance.

### DermImpact

New York, NY

*Founding Engineer*

February 2026 - May 2026

- Developed a full-stack platform (**Next.js**, **TypeScript**) for AI-assisted dermatology notes and image analysis, validated in a live pilot with a **30+ physician dermatology clinic**.
- Architected HIPAA-oriented data pipelines with **RLS**, encrypted image storage, and **PostHog** usage analytics.

### NYU Biokind Analytics

New York, NY

*Data Engineer*

January 2026 - May 2026

- Engineered **ETL pipelines** to process **35k+** nonprofit donor engagement records, improving platform engagement by **20%**.
- Architected a **FastAPI** system to orchestrate pipelines and automate **database maintenance** via API integrations, reducing manual data processing time by **60%** and accelerating data delivery to organizers.

### Pulse Foundry

New York, NY

*Software Engineer*

December 2025 - May 2026

- Built and deployed an automated newsletter pipeline (**FastAPI**, **Supabase**, **OpenAI**) with **concurrent ingestion**, **semantic ranking**, and **scheduled workers**, serving **3,000+ subscribers**.
- Developed an internal **React** dashboard enabling editors to review, filter, and configure pipeline parameters in real time.

## Projects

### HyperZero | Python, PyTorch, FastAPI, React, Three.js

May 2026 - June 2026

- Built an **RL agent** for multidimensional Connect-K combining **MCTS**, self-generated data, and policy-value neural networks.
- Achieved **87% mean win rate** against tactical, heuristic, and MCTS baselines across multiple board configurations.
- Developed a training pipeline optimized for high-throughput with **batched inference** and **parallelized simulations**.
- Engineered a generalized environment for arbitrary board configurations, enabling experiments across multiple game variants.
- Deployed a real-time 3D game interface with **WebSocket**-based agent API and live move visualizations.

### Switchboard | Go, SQLite, Docker, Google Cloud, Lambda Cloud

April 2026 - June 2026

- Developed a multi-cloud ML orchestration platform for managing training and inference workloads across local and cloud compute environments, with provider abstractions for **remote execution**, **artifact storage**, and **workload routing**.
- Implemented **concurrent job execution** and **lifecycle management** systems to support scalable ML workflows.

### Neural-ViT | Python, PyTorch, Vertex AI, BigQuery, Google Cloud Storage

December 2025 - January 2026

- Implemented a **3D Vision Transformer** for genotype classification on multi-trial neural spectrograms, achieving **0.96 AUC** (+9% over baseline models), trained on NVIDIA A100s via Google Cloud Vertex AI.
- Built an end-to-end ML pipeline processing **77k+ trials** with preprocessing, checkpointing, experiment tracking, and configurable hyperparameter sweeps with optimization techniques such as **mixed precision** and **learning rate scheduling**.

### AdaHealth - 1st Place, Pulse NYC AI Hackathon (20 teams) | React, Node.js, MongoDB, OpenAI

December 2025

- Built an AI health companion that conducts elderly patient check-ins with **real-time voice** and **LLM-based safety scoring**.
- Developed a companion dashboard enabling caretakers to track patient health and conversation metrics across sessions.